

Unbiased estimation via randomization - Application to Option Pricing

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(Q1)

1.

$$\begin{aligned}\|Y_\ell - Y\|_2^2 &= E[(Y_\ell - Y)^2] \\ &= E[Y_\ell^2 - 2Y_\ell Y + Y^2] \\ &= E[Y_\ell^2] - 2E[Y_\ell]E[Y] + E[Y^2] \quad \text{by independence of } Y_\ell \text{ and } Y \\ &\rightarrow E[Y^2] - 2E[Y]^2 + E[Y^2] \\ &= 2 \underbrace{(E[Y^2] - E[Y]^2)}_{\text{Var}[Y]} \\ &= 2\text{Var}[Y] \not\rightarrow 0 \quad \text{if } \text{Var}[Y] \neq 0\end{aligned}$$

By assumption, N is almost surely finite, i.e.

$$P(N = \infty) = 0.$$

But

$$\{N \geq 1\} \supset \{N \geq 2\} \supset \{N \geq 3\} \supset \dots$$

and $\cap_{\ell \geq 1} \{N \geq \ell\} = \{N = \infty\}$. so by continuity of probability,

$$0 = P(N = \infty) = \lim_{\ell \rightarrow \infty} P(N \geq \ell),$$

so for ℓ big enough

$$\frac{\|Y_{\ell-1} - Y\|_2^2}{P(N \geq \ell)} \geq \frac{\text{Var}[Y]}{P(N \geq \ell)}.$$

But since $P(N \geq \ell) \rightarrow 0$ we have

$$\frac{\text{Var}[Y]}{P(N \geq \ell)} \rightarrow \infty$$

so

$$\sum_{\ell=1}^{\infty} \frac{\|Y_{\ell-1} - Y_\ell\|_2^2}{P(N \geq \ell)} = \infty$$

2.

We want to show

$$\frac{\|Y_{\ell-1} - Y\|_2^2}{P(N \geq \ell)} < \infty. \quad (1)$$

By the strong order p assumption and the fact that we have a geometric time step, we have

$$\begin{aligned} \frac{\|Y_{\ell-1} - Y\|_2^2}{P(N \geq \ell)} &\leq \frac{\left(C \left(\frac{T}{2^{\ell-1}}\right)^p\right)^2}{q^\ell} \\ &= \underbrace{CT^{2p}2^{2p}}_{\tilde{C}} \left(\frac{1}{2^{2p}q}\right)^\ell. \end{aligned}$$

In order for this geometric series to converge, we need $\frac{1}{2^{2p}q} < 1$, i.e. $q > 2^{-2p}$.

We now proceed to show that $\bar{Z}$ has finite variance. By definition, $\text{Var}[\bar{Z}] = E[\bar{Z}^2] - [E\bar{Z}]^2$. But by Jensen's inequality, we have

$$E[\bar{Z}] = E[Y] \leq (E[Y^2])^{1/2} < \infty \quad \text{because } Y \in L^2(\Omega).$$

So we just need to show $E[\bar{Z}^2] < \infty$. But

$$\begin{aligned} E[\bar{Z}^2] &= \sum_{\ell=0}^{\infty} \frac{\bar{\nu}_\ell}{P(N \geq \ell)} \\ &= \sum_{\ell=0}^{\infty} \frac{\|Y_{\ell-1} - Y\|_2^2 - \|Y_\ell - Y\|_2^2}{P(N \geq \ell)} \\ &\leq \sum_{\ell=0}^{\infty} \frac{\|Y_{\ell-1} - Y\|_2^2}{P(N \geq \ell)} < \infty \quad \text{by 1} \end{aligned}$$

(Q2)

1.

$$\begin{aligned} E[\bar{\tau}] &= E\left[\sum_{j=1}^N \bar{t}_j\right] \\ &= E\left[\sum_{j=1}^{\infty} \bar{t}_j \mathbb{1}(N \geq j)\right] \\ &= \sum_{j=1}^{\infty} \bar{t}_j E[\mathbb{1}(N \geq j)] \quad \text{by Tonelli} \\ &= \sum_{j=1}^{\infty} \bar{t}_j P(N \geq j) \end{aligned}$$

2.

$$\min_F \underbrace{\left(\sum_{j=0}^{\infty} \frac{\beta_j}{F_j}\right)}_{:=A(F)} \underbrace{\left(\sum_{j=0}^{\infty} t_j F_j\right)}_{:=B(F)}$$

is equivalent to solving

$$\begin{aligned} \min_F \quad & \sum_{j=0}^{\infty} \frac{\beta_j}{F_j} \\ \text{subject to} \quad & \sum_{j=0}^{\infty} t_j F_j = C \end{aligned}$$

indeed if F^* is a minimizer of $A(F)B(F)$, then F^* is a minimizer of

$$\min_F A(F) \quad \text{s.t.} \quad B(F) = \underbrace{B(F^*)}_{:=C}.$$

Indeed, if this is not the case, there exists $\hat{F}$ such that $A(\hat{F}) < A(F^*)$ with $B(\hat{F}) = B(F^*)$. But then $A(\hat{F})B(\hat{F}) < A(F^*)B(F^*)$ which contradicts optimality of F^* .

This is a constrained optimization problem so we use Lagrange multipliers:

$$\mathcal{L}(F, \lambda) = \sum_{j=0}^{\infty} \frac{\beta_j}{F_j} + \lambda \left(\sum_{j=0}^{\infty} t_j F_j - C \right)$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial F_i} &= 0 \\ \frac{-\beta_i}{F_i^2} + \lambda t_i &= 0 \\ F_i &= \frac{1}{\sqrt{\lambda}} \sqrt{\frac{\beta_i}{t_i}} \end{aligned}$$

In order that $F_0 = 1$, we need

$$\frac{1}{\sqrt{\lambda}} \sqrt{\frac{\beta_0}{t_0}} = 1$$

so $\lambda = \beta_0/t_0$, i.e.

$$F_i = \frac{\sqrt{\beta_i/t_i}}{\sqrt{\beta_0/t_0}} =: F_i^*.$$

So if F^* is feasible (as stated in the assumption of the question) we have that F^* satisfies the KKT conditions. Moreover, the problem is strictly convex so F^* is the unique optimal solution.

3.

$$\begin{aligned} \|\Delta_j\|_2 &= \|Y_j - Y_{j-1}\| \\ &\leq \underbrace{\|Y_j - Y\|_2}_{C2^{-pj}} + \underbrace{\|Y - Y_{j-1}\|_2}_{C2^{-p(j-1)}} \\ &\leq (C + C2^p)2^{-pj} \end{aligned}$$

so

$$\begin{aligned} E \left[\Delta_j^2 \right] &= \|\Delta_j\|_2^2 \\ &= \underbrace{(C + C2^p)^2}_{\tilde{C}} 2^{-2pj} \\ &= O(2^{-2pj}) \end{aligned}$$

4.

In order to get the best balance between the variance and the cost, we note that we need to consider the optimal $F_j^* = P(N \geq j)$.

We know $F_j^* \propto \sqrt{\frac{\nu_j}{t_j}} \propto \sqrt{\frac{2^{-2pj}}{2^j}} = 2^{-j(p+\frac{1}{2})}$. Since we assumed that the survival distribution is of the form $P(N \geq \ell) = 2^{-r\ell}$, we have $r^* = p + \frac{1}{2}$ is balancing the variance and the cost.

Let us determine the conditions for $E[\bar{\tau}]$ and $\text{Var}[\bar{Z}]$ to be finite.

$$\begin{aligned} E[\bar{\tau}] &= \sum_{j=0}^{\infty} t_j P(N \geq j) \\ &\leq \sum_{j=0}^{\infty} c 2^j 2^{-rj} \\ &= \sum_{j=0}^{\infty} C 2^{j(1-r)} \end{aligned}$$

In order for the series to converge, we need $j(1-r) < 0$, i.e. $r > 1$.

By the same argument as in (Q1.2), $\text{Var}[\bar{Z}] < \infty$ is equivalent to $E[\bar{Z}^2] < \infty$. But

$$\begin{aligned} E[\bar{Z}^2] &= \sum_{\ell=0}^{\infty} \frac{\bar{\nu}_\ell}{P(N \geq \ell)} \\ &= \sum_{\ell=0}^{\infty} \frac{C 2^{-2p\ell}}{2^{-r\ell}} \\ &= \sum_{\ell=0}^{\infty} C 2^{\ell(r-2p)} \end{aligned}$$

and in order for the series to converge, we need $j(r-2p) < 0$, i.e. $r < 2p$.

4.

With the optimal choice $r^* = p + \frac{1}{2}$, the two finiteness conditions become

$$r^* > 1 \iff p > \frac{1}{2}, \quad r^* < 2p \iff p > \frac{1}{2}.$$

Both are equivalent and require $p > \frac{1}{2}$. For Euler–Maruyama ($p = \frac{1}{2}$) the optimal rate is $r^* = 1$, which sits exactly on the boundary: the series $E[\bar{\tau}] = \sum_j t_j 2^{-j} = \sum_j c \cdot 2^j \cdot 2^{-j} = \sum_j c$ diverges. Hence EM does not yield a finite expected computational cost, regardless of the choice of q .

Milstein ($p = 1$) gives $r^* = \frac{3}{2}$, comfortably inside both conditions ($1 < \frac{3}{2} < 2$), so $E[\bar{\tau}] < \infty$ and $\text{Var}(\bar{Z}) < \infty$.

Increasing p beyond 1 further improves efficiency: the work–variance product scales as $E[\bar{\tau}] \cdot \text{Var}(\bar{Z}) \propto (r^* - 1)^{-1} (2p - r^*)^{-1} = (p - \frac{1}{2})^{-2}$, which decreases as p increases. However the gain is large when p is close to $\frac{1}{2}$ (EM to Milstein) and diminishes for $p \gg 1$. In practice, going from $p = \frac{1}{2}$ to $p = 1$ is a qualitative improvement (from infeasible to feasible), while going from $p = 1$ to higher orders yields only a quantitative, diminishing gain.

(Q3)

Q3.1 – Milstein scheme with coupling between levels

At level ℓ , the Milstein scheme uses $n_\ell = 2^\ell$ steps of size $h_\ell = T/2^\ell$. Fine Brownian increments $\Delta B_k^{(\ell)} \sim \mathcal{N}(0, h_\ell)$ are generated for $k = 0, \dots, 2^\ell - 1$, and the coarse increments at level $\ell - 1$ are obtained by the *Brownian refinement coupling*:

$$\Delta B_k^{(\ell-1)} := \Delta B_{2k}^{(\ell)} + \Delta B_{2k+1}^{(\ell)}, \quad k = 0, \dots, 2^{\ell-1} - 1.$$

Both levels are then driven by the *same* Brownian path. The fine and coarse Milstein approximations of S_T read

$$\begin{aligned} S_{k+1}^f &= S_k^f \left(1 + \mu h_\ell + \sigma \Delta B_k^{(\ell)} + \frac{\sigma^2}{2} ((\Delta B_k^{(\ell)})^2 - h_\ell) \right), \\ S_{k+1}^c &= S_k^c \left(1 + \mu h_{\ell-1} + \sigma \Delta B_k^{(\ell-1)} + \frac{\sigma^2}{2} ((\Delta B_k^{(\ell-1)})^2 - h_{\ell-1}) \right). \end{aligned}$$

The level difference $\Delta_\ell = \varphi(S_T^f) - \varphi(S_T^c)$ is small because both paths are driven by consistent increments, which is essential for reducing the variance of the coupled-sum estimator.

Why Milstein rather than Euler–Maruyama? As shown in Q2.4, the coupled-sum estimator requires a numerical scheme with strong order $p > \frac{1}{2}$ for both $E[\bar{\tau}]$ and $\text{Var}(\bar{Z})$ to be finite at the optimal truncation rate $r^* = p + \frac{1}{2}$. Euler–Maruyama has strong order $p = \frac{1}{2}$, placing it exactly on the boundary where the expected computational cost $E[\bar{\tau}]$ diverges. Milstein achieves $p = 1$, giving $r^* = \frac{3}{2}$ and ensuring both quantities are finite with margin. This makes Milstein the minimal scheme for which the coupled-sum estimator is well-defined with the choice $q = 2^{-3/2}$.

Q3.2 – Coupled-sum estimator

We implement the coupled-sum unbiased estimator (4) using the Milstein scheme for the GBM dynamics (15), with truncation level N drawn from a geometric distribution $\mathbb{P}(N \geq \ell) = q^\ell$ and $q = 2^{-3/2}$ (the optimal rate from Q2.4 with strong order $p = 1$). The results over $R = 10$ independent runs are reported in Table 1 for budgets $c \in \{20\,000, 100\,000, 500\,000\}$.

Budget	Mean	Variance	Bias	RMSE	95% CI
20 000	0.105445	2.22×10^{-2}	9.39×10^{-4}	3.56×10^{-3}	[0.09974, 0.11115]
100 000	0.104987	2.20×10^{-2}	4.81×10^{-4}	1.32×10^{-3}	[0.10232, 0.10766]
500 000	0.104607	2.18×10^{-2}	1.02×10^{-4}	8.25×10^{-4}	[0.10346, 0.10575]

Table 1: Coupled-sum estimator results (true value $\alpha = 0.104506$, $R = 10$ runs).

The estimator is unbiased: the empirical bias is negligible at all budgets and decreases with the budget. The RMSE and CI width both shrink roughly as $c^{-1/2}$, consistent with the central limit theorem (9).

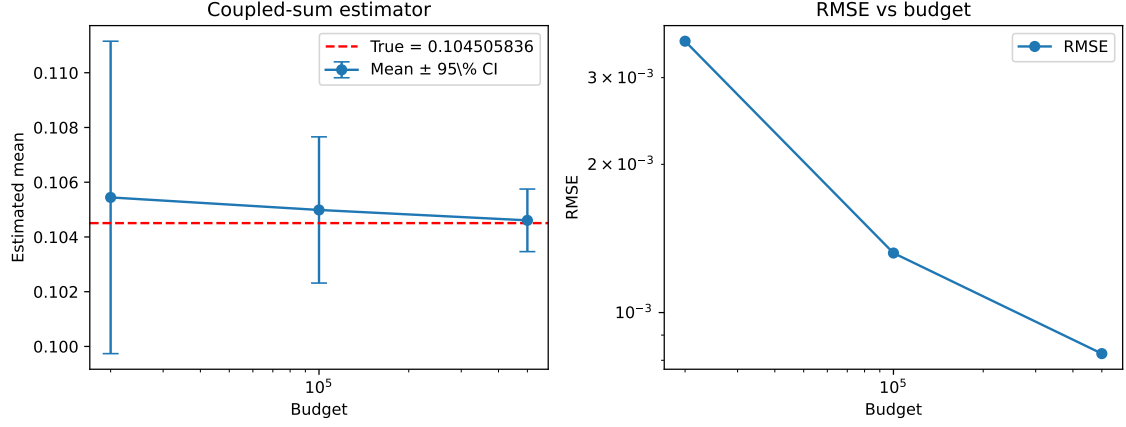


Figure 1: Left: estimated mean with 95% CI as a function of budget. Right: RMSE vs budget.

Q3.3 – Comparison with standard Monte Carlo

We compare the coupled-sum and standard MC estimators at the same budgets. For standard MC we balance discretization and statistical errors by choosing $h \approx c^{-1/3}$ and $N \approx c^{2/3}$ (weak order $\zeta = 1$ for Milstein). Results are reported in Tables 2 and 3.

Budget	Mean	Variance	Bias	RMSE	Work $\times$ Var	95% CI
20 000	0.104539	2.18×10^{-2}	3.3×10^{-5}	2.41×10^{-3}	1.72×10^{-1}	[0.09877, 0.11031]
100 000	0.104194	2.17×10^{-2}	-3.1×10^{-4}	1.66×10^{-3}	1.73×10^{-1}	[0.10162, 0.10677]
500 000	0.104688	2.19×10^{-2}	1.8×10^{-4}	4.34×10^{-4}	1.68×10^{-1}	[0.10355, 0.10582]

Table 2: Coupled-sum estimator ($R = 10$ runs).

Budget	Mean	Variance	Bias	RMSE	Work $\times$ Var	95% CI
20 000	0.102961	2.06×10^{-2}	-1.54×10^{-3}	5.38×10^{-3}	5.57×10^{-1}	[0.09260, 0.11332]
100 000	0.105194	2.16×10^{-2}	6.9×10^{-4}	2.33×10^{-3}	9.92×10^{-1}	[0.09899, 0.11140]
500 000	0.105549	2.19×10^{-2}	1.04×10^{-3}	2.23×10^{-3}	1.73×10^0	[0.10189, 0.10921]

Table 3: Standard MC estimator ($R = 10$ runs).

Several observations follow from these results.

- **Bias.** The coupled-sum estimator is unbiased by construction; its empirical bias is negligible at all budgets. Standard MC retains a persistent discretization bias of order 10^{-3} , visible even at budget 500 000, which causes the RMSE to plateau rather than decrease proportionally to $c^{-1/2}$.
- **Work–variance product.** For the coupled-sum estimator, $\mathbb{E}[\bar{\tau}] \cdot \text{Var}(\bar{Z})$ stays roughly constant (≈ 0.17) across budgets, consistent with the asymptotic result (9). For standard MC it grows as $c^{1/3}$: at budget 500 000 it is an order of magnitude larger, confirming the superior asymptotic efficiency of the coupled-sum estimator.
- **Confidence intervals.** At budget 500 000 the coupled-sum CI width is roughly half that of standard MC, reflecting its lower RMSE.

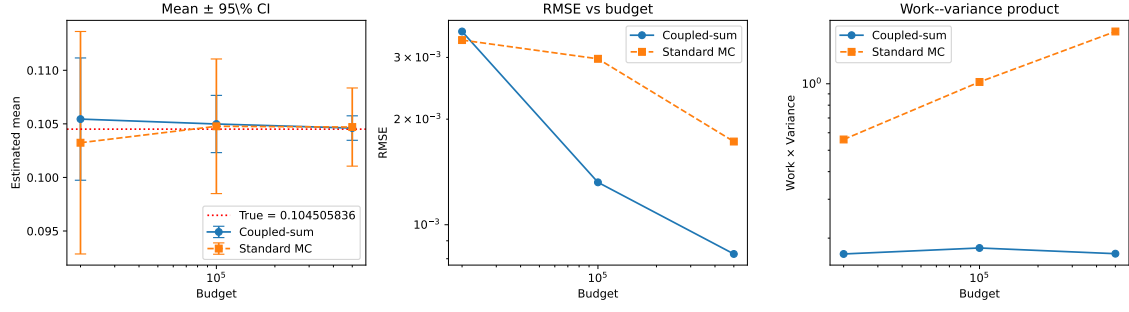


Figure 2: Left: mean with 95% CI. Centre: RMSE vs budget. Right: work-variance product. Coupled-sum (circles) vs standard MC (squares).

(Q4)

1.

Suppose by contradiction that there is a path with $\tau_\epsilon < \infty$. Then by continuity of V_t there is a time where V_t first crosses 0. At this point the process reads $dV_t = \kappa\theta > 0$ which pushes the process back to a positive value. So the process can never be negative. But then the original process is also always positive.

2.

The idea is to find a Lyapunov function $f(x)$ which explodes when $x \rightarrow 0$, for example $f(x) = x^{-m}$. In order to derive the dynamics of $f(V_t)$ we use Itô's formula:

$$df(V_t) = f'(V_t)dV_t + \frac{1}{2}f''(V_t)\underbrace{d\langle V \rangle_t}_{\sigma^2 V_t dt}$$

with

$$\begin{aligned} f'(x) &= -mx^{-m-1} \\ f''(x) &= m(m+1)x^{-m-2} \end{aligned}$$

So we have

$$\begin{aligned} df(V_t) &= -mV_t^{-m-1}dV_t + \frac{1}{2}m(m+1)V_t^{-m-2}\sigma^2 V_t dt \\ &= -mV_t^{-m-1}\kappa(\theta - V_t)dt - mV_t^{-m-1}\sigma\sqrt{V_t}dB_t + \frac{1}{2}m(m+1)V_t^{-m-1}\sigma^2 dt \\ &= \left[-mV_t^{-m-1}\kappa(\theta - V_t) + \frac{1}{2}m(m+1)V_t^{-m-1}\sigma^2 \right] dt \\ &\quad - mV_t^{-m-1}\sigma\sqrt{V_t}dB_t \\ &= m\kappa V_t^{-m}dt - mV_t^{-m-\frac{1}{2}}\sigma dB_t \end{aligned}$$

where in the last line we set $m = 2\kappa\theta/\sigma^2 - 1$ in order to simplify the expression.

Now it is important to note that we cannot apply Itô on all of $[0, t]$ because V_t^{-m} blows up if V_t hits 0. But we can write Itô's formula for the stopped process

$$V_{t \wedge \tau_\epsilon} \quad \text{where } \tau_\epsilon = \inf\{t : V_t \leq \epsilon\}.$$

We obtain

$$V_{t \wedge \tau_\epsilon}^{-m} = V_0^{-m} + \int_0^t m\kappa V_{s \wedge \tau_\epsilon}^{-m} ds - \int_0^t m V_{s \wedge \tau_\epsilon}^{-m-1/2} \sigma dB_s.$$

Now taking expectation, the term with the Brownian motion disappears because it is a martingale since

$$E\left[\int_0^{t \wedge \tau_\epsilon} V_s^{-2(m+1/2)} ds\right] \leq \epsilon^{-2m-1} t < \infty$$

so

$$E[V_{s \wedge \tau_\epsilon}^{-m}] = V_0^{-m} + m\kappa \int_0^t E[V_{s \wedge \tau_\epsilon}^{-m}] ds$$

where we passed the expectation inside the integral using Tonelli's theorem. By Grönwall's inequality, we then get

$$E[V_{s \wedge \tau_\epsilon}^{-m}] \leq V_0^{-m} e^{m\kappa t}.$$

Now note

$$\begin{aligned} P(\tau_\epsilon < t) &= P(V_{s \wedge \tau_\epsilon}^{-m} \geq \epsilon^{-m}) \\ &\leq E[V_{s \wedge \tau_\epsilon}^{-m}] \quad \text{by Markov's inequality} \\ &\leq V_0^{-m} e^{m\kappa t} \epsilon^m. \end{aligned}$$

We now have to show

$$P(V_t > 0 \forall t \in [0, T]) = 1.$$

This is because

$$\begin{aligned} P(\exists t \in [0, T] : V_t \leq 0) &\leq P(\tau_\epsilon \leq T) \\ &\leq V_0^{-m} e^{m\kappa T} \epsilon^m \rightarrow 0 \quad \text{when } \epsilon \rightarrow 0 \end{aligned}$$

So

$$P(\exists t \in [0, T] : V_t \leq 0) = 0$$

and thus

$$P(V_t > 0 \forall t \in [0, T]) = 1$$

3.

Again, we use Itô's formula with

$$\begin{aligned} g(x) &= x^{1/2} \\ g'(x) &= \frac{1}{2} x^{-1/2} \\ g''(x) &= -\frac{1}{4} x^{-3/2} \end{aligned}$$

because

$$Z_t = \sqrt{V_t} = g(V_t)$$

. We get

$$\begin{aligned}
dZ_t &= dg(V_t) \\
&= g'(V_t)dV_t + \frac{1}{2}g''(V_t)\underbrace{d\langle V \rangle_t}_{\sigma^2 V_t dt} \\
&= \frac{1}{2}V_t^{-1/2}dV_t - \frac{1}{8}V_t^{-3/2}\sigma^2 V_t dt \\
&= \frac{1}{2}V_t^{-1/2}\kappa(\theta - V_t)dt + \frac{1}{2}V_t^{-1/2}\sigma V_t^{1/2}dB_t - \frac{1}{8}V_t^{-1/2}\sigma^2 dt \\
&= V_t^{-1/2}\left[\frac{1}{2}\kappa(\theta - V_t) - \frac{1}{8}\sigma^2\right]dt + \frac{1}{2}\sigma dB_t \\
&= \left[\frac{1}{2}\kappa(\theta V_t^{-1/2} - V_t^{1/2}) - \frac{1}{8}\sigma^2 V_t^{-1/2}\right]dt + \frac{1}{2}\sigma dB_t \\
&= \frac{1}{2}\kappa\left[\left(\theta - \frac{\sigma^2}{4\kappa}\right)Z_t^{-1} - Z_t\right]dt + \frac{1}{2}\sigma dB_t
\end{aligned}$$

(Q5)

Q5.2 – Option pricing under the CIR model

We consider a European call-type payoff $Y = \varphi(V_T)$ with $\varphi(v) = e^{-\mu T}(v - K)^+$, $\mu = 0.05$, $K = 0.25$, discretized with the drift-implicit Milstein scheme (25). Since no analytical formula is available, we estimate the reference value using a large-budget ($c = 5\,000\,000$) coupled-sum run, giving $\alpha_{\text{ref}} \approx 0.006370$. Tables 4 and 5 report results over $R = 10$ runs for budgets $c \in \{1\,000, 10\,000, 50\,000\}$.

Budget	Mean	Variance	Bias	RMSE	Work×Var	95% CI
1 000	0.007449	2.18×10^{-3}	1.08×10^{-3}	4.38×10^{-3}	1.52×10^{-2}	[0.000383, 0.014516]
10 000	0.006910	1.83×10^{-3}	5.40×10^{-4}	9.91×10^{-4}	1.38×10^{-2}	[0.004630, 0.009190]
50 000	0.006080	1.61×10^{-3}	-2.90×10^{-4}	5.09×10^{-4}	1.32×10^{-2}	[0.005087, 0.007073]

Table 4: CIR coupled-sum estimator ($R = 10$ runs, reference $\alpha_{\text{ref}} \approx 0.006370$).

Budget	Mean	Variance	Bias	RMSE	Work×Var	95% CI
1 000	0.005696	5.18×10^{-4}	-6.75×10^{-4}	2.78×10^{-3}	5.18×10^{-3}	[0.001472, 0.009920]
10 000	0.006265	7.40×10^{-4}	-1.05×10^{-4}	1.04×10^{-3}	1.63×10^{-2}	[0.003809, 0.008722]
50 000	0.005976	7.42×10^{-4}	-3.94×10^{-4}	5.46×10^{-4}	2.74×10^{-2}	[0.004532, 0.007420]

Table 5: CIR standard MC estimator ($R = 10$ runs).

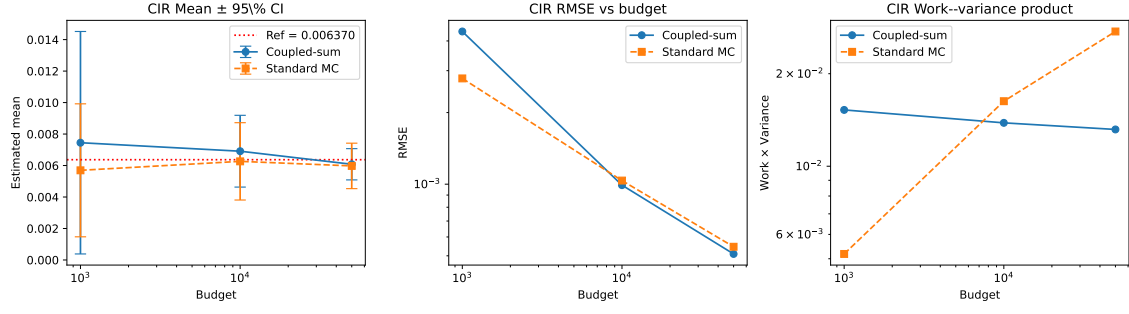


Figure 3: CIR option pricing. Left: mean with 95% CI. Centre: RMSE vs budget. Right: work-variance product. Coupled-sum (circles) vs standard MC (squares).

The qualitative picture mirrors the GBM case. The coupled-sum estimator remains unbiased while standard MC retains a persistent discretization bias. A key difference is that both methods now have comparable work-variance products at small budgets: at $c = 1\,000$ the coupled-sum product (1.52×10^{-2}) is slightly larger than standard MC (5.18×10^{-3}), reflecting the overhead of the multilevel correction at very small budgets. At larger budgets ($c = 50\,000$) the trend reverses: the work-variance product of standard MC grows as $c^{1/3}$ while that of the coupled-sum stays roughly constant, confirming the superior asymptotic efficiency of the unbiased estimator.